AI in Finance Certificate

Cornell University®

Preparing Financial Leaders for an AI-Driven World

The Economics of AI in Finance

April 20, 2026

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About Me

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Background

PhD, University of Chicago
Previously: Assistant Professor of Finance, Swiss Finance Institute
Research Associate, Columbia Business School (Real Estate)
Investment banking analyst, boutique advisory

Research interests

- How financial frictions affect firm decisions to adopt new technology
- The trade-off between production and innovation: running the current system vs. investing in something new
- External financing costs, competition, and spillovers

My Role in This Program

What I'll cover

- Risk, frictions, equilibrium dynamics
- Where AI changes the underlying economics—and where it doesn't

What I won't cover

- Which specific applications will succeed
- How to implement them—that's the rest of the program

My approach

- Finance first, AI second
- Learn from past technological changes in intermediation
- A framework for evaluating AI claims honestly

Two Lectures

Lecture 1 *Risk, frictions, prediction*

- What financial markets are for
- What makes them hard (and where AI changes the economics)
- Where prediction creates value—and where it doesn't

Lecture 2 *Equilibrium, agents, new markets*

- What happens when everyone adopts the same tools
- AI as an economic agent
- New markets that become possible

Devil's advocate Not skepticism—stress-testing. The best way to evaluate a technology is to push on it.

What Financial Markets Are For

Financial markets facilitate the transfer of resources across time, between people, under uncertainty.

Measure risk *broken by: information asymmetry*

- The finance question: estimate default probabilities, correlations, tail exposures
- The AI pitch today: Upstart, Zest AI score borrowers FICO can't see; LLMs parse earnings calls for hidden risk

Transfer risk *broken by: moral hazard*

- The finance question: hedging, securitization, insurance—moving risk changes behavior
- The AI pitch today: telematics at Progressive and Tesla price behavior in real time, not just observe it

Price risk *broken by: microstructure & transaction costs*

- The finance question: bid-ask spreads, price impact, adverse selection
- The AI pitch today: Citadel Securities and Virtu use ML to flag toxic flow; MarketAxess matches corporate bond trades dealers won't warehouse

Risk and Return

Risk and Return: The CAPM

Sharpe (1964); Lintner (1965)

The foundational question of asset pricing: why do some assets earn higher returns than others?

CAPM: only systematic risk is compensated

$$E(R_i) = R_f + \beta_i \left[E(R_m) - R_f\right]$$

- β: how the asset moves with the market. High β = compensated for more risk.
- R_f: the risk-free rate; $E(R_m) - R_f$: the equity risk premium
- Idiosyncratic risk is diversifiable and earns nothing

Alpha: what the model cannot explain

- Empirically: $R_i = \alpha_i + \beta_i R_m + \varepsilon_i$
- $\alpha > 0$ means the asset earns more than its risk justifies
- The holy grail of active management—and a sign the model is missing something

Sixty Years of Searching for Alpha

Fama & French (1993); Harvey, Liu & Zhu (2016)

If CAPM were right, α would be zero. It is not. The field has spent sixty years finding things CAPM missed.

The progression

- **1964:** CAPM—one factor (market beta)
- **1993:** Fama-French—three factors (market, size, value): small and cheap stocks earn more than CAPM predicts
- **1997:** Carhart adds momentum—four factors
- **2000s:** Hundreds of published "anomalies"—each claiming to explain some of α

The problem (Harvey, Liu & Zhu 2016)

- Of $\sim$300 published factors, most do not survive proper multiple-testing correction
- Many are artifacts of data mining, not real compensation for risk
- The field does not know which factors are signal and which are noise

What Neural Networks Found

Gu, Kelly & Xiu (2020)

Gu, Kelly & Xiu ran every major ML method on 50 years of US stock returns. Neural networks won.

The result

- Out-of-sample $R^2 \approx 0.4\%$ per month at the stock level
- That sounds small. Economically, it produces Sharpe ratios comparable to professional quant funds.
- What the net found: nonlinear interactions that linear factor models systematically miss (e.g., momentum behaves differently in high-vs-low-volatility regimes)

The ambiguity

- Interpretation 1: **investable alpha**—a market inefficiency that a fund could trade
- Interpretation 2: **unmeasured risk**—compensation for a risk factor we didn't know to price
- The data cannot distinguish the two. Both interpretations fit.

This ambiguity is central to Lecture 2: if it is alpha, adoption should compete it away. If it is risk compensation, it should persist.

Where AI Enters the Argument

Prediction meets asset pricing

The Debate AI Is Walking Into

Asset pricing has argued for fifty years over a single question: is this return risk compensation or market inefficiency? The answer determines whether a strategy should persist or be competed away.

The long-standing debate

- Fama: markets are efficient; returns above beta reflect unmeasured risk
- Shiller, Thaler: markets are not always efficient; some returns reflect systematic mispricing
- Every "factor" in the 400-factor zoo carries the same ambiguity GKX shows at the ML frontier

What AI could actually do

- **(1) Disentangle risk from inefficiency.** Richer, nonlinear models let us test whether a factor's payoff correlates with states of the world (risk) or with investor attention and frictions (inefficiency).
- **(2) Change the nature of inefficiency itself.** When AI agents make markets more informationally efficient in some dimensions, they may create new inefficiencies in others — coordination, reflexivity, emergent crowding. This is Lecture 2.

The right question is not "can AI beat the market?" It is "does AI change which question asset pricing is asking?"

Why Risk Is Hard — And Where AI Can Help

Where AI might break through

Fat tails

- 2008 was a 25σ event under Goldman's models
- ML fits flexible distributions—but tail events are rare in training

Correlation breaks

- LTCM 1998: stable correlations broke; $4.6B lost in weeks
- AI models regime-switching; structural shifts are unseen by definition

Where AI likely cannot

Model uncertainty

- VaR assumed normality until 2008
- AI adds models; it cannot resolve which is right

Incomplete markets

- No futures for firm-specific revenue; the *contracts* don't exist
- AI measures what is tradable; it cannot create the market

LLMs are fundamentally about prediction. Prediction is a real capability — but the frictions finance is built to solve are not prediction problems. That's the gap we need to understand next.

The Frictions That Shape Finance

Why prediction alone is not enough

The Finance Problem Is Not Just Prediction

We have risk (which is fundamentally hard) and prediction (which has its own limits). But the bigger problem is the environment we are predicting in.

The economic environment itself is shaped by frictions

- Borrowers know things lenders do not
- Securitizers stop screening once they transfer risk
- Market makers widen spreads when they suspect private information

Each friction breaks a specific risk function

- Information asymmetry $\rightarrow$ breaks *measurement*
- Moral hazard $\rightarrow$ breaks *transfer*
- Microstructure $\rightarrow$ breaks *pricing*

This is where AI meets the real test: for AI to change the economics, it has to address one of these frictions—not just predict faster within them.

Three Frictions, Three AI Pitches You Will See

Information asymmetry makes you price what you cannot measure. Moral hazard makes the transfer change behavior. Microstructure makes even the pricing mechanism costly. Each friction has a vendor pitching an AI solution right now.

1 Information asymmetry — *breaks risk measurement*

The pitch: Upstart, Zest AI score thin-file borrowers; LLMs read loan apps and earnings calls.

The test: Does new information reach the right people, or just change who gets sorted where?

2 Moral hazard and enforcement — *breaks risk transfer*

The pitch: Progressive Snapshot, Tesla Insurance, dynamic covenants—AI monitors behavior continuously.

The test: Does the contract change, or are we just watching the same behavior more carefully?

3 Microstructure and transaction costs — *breaks risk pricing*

The pitch: Citadel Securities, Virtu, MarketAxess—AI finds liquidity and prices toxic flow.

The test: Real gains—but how fast do competitors reverse-engineer the signal?

Information Asymmetry → Risk Measurement

Akerlof (1970)

FICO (1989) and what AI claims to add (now)

- FICO solved this for borrowers with credit history—subprime auto lending still blends 4% prime and 18% deep subprime
- 45M Americans have no FICO score at all—the asymmetry persists where it matters most
- Upstart, Zest AI, Pagaya: use unstructured data (rent, utilities, employment text) to score the thin-file population

Worked Example: A Lending Pool

1,000 applicants

600 low-risk default 3% fair rate 5%	**400 high-risk** default 15% fair rate 18%

Lender cannot distinguish → blended rate: 10%

↓

300 leave get 5–7% elsewhere	**300** low-risk	**400** high-risk

New blended default: 10% → rate rises to 13% → spiral

With AI scoring: low-risk get 5% (stay) · high-risk get 18% (priced accurately) · $30M in new lending

But: if AI scores are less accurate for some demographic groups, the gains are not equally distributed.

What AI Changes About Information Asymmetry

Fuster et al. (2022)

AI reveals information the borrower used to hold privately. This improves the pool—but changes who gets sorted where.

Where AI breaks through

New signals, new borrowers

- LLMs process rent, utility, and application text at scale
- Reaches the 45M Americans without a FICO score
- Fuster: ML approves ∼8% more borrowers at equal default risk

Where it cannot

The distribution of who benefits

- New signals correlate with race, geography, and wealth
- The same model helps one group and hurts another at *any* threshold
- This is not a bug to be fixed—it is a design choice about whose welfare to maximize

Better prediction does not equal better outcomes for everyone.

ML Underwriting: The Evidence

Fuster et al. (2022), Figure 1

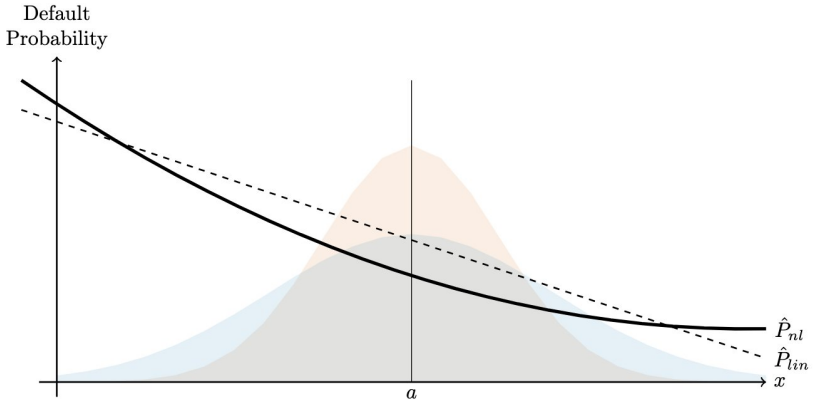


Figure 1. Unequal Effects of Better Technology.

Reading this figure

- Solid line: ML prediction of default
- Dashed line: logistic regression prediction
- The two curves cross—ML is higher for some borrowers, lower for others
- Shaded areas: two demographic groups, distributed differently along the risk score
- **At any threshold *a*, switching from logit to ML changes who is approved—and the effect differs by group**

Moral Hazard → Risk Transfer

Jensen & Meckling (1976); Hart & Moore (1998)

Transferring risk changes the behavior of the party who no longer bears it. This is an incentive problem, not an information problem—which means AI monitoring, by itself, does not solve it.

Then: Countrywide (2008)

- Securitized >90% of originated mortgages; screening had no private benefit
- Keys et al. (2010): loans just above the 620 FICO cutoff defaulted 10–25% more often

Now: private credit and covenant-lite (2020s)

- $1.7T private credit market; covenant-lite is now >90% of leveraged loans
- AI monitoring platforms (Rubik, Numerated) parse covenant docs and flag breaches in real time
- But the covenants themselves are weak—monitoring something toothless more carefully does not help

The structural fix, when it works: Dodd-Frank's 5% risk retention forced skin in the game. That realigned incentives. Better monitoring alone never has.

Worked Example: The Securitization Cutoff

Keys, Mukherjee, Seru & Vig (2010)

A natural experiment: same borrowers, different incentives, different outcomes.

FICO 620 cutoff

FICO 619	Economically identical borrowers	FICO 621
Hard to securitize Originator keeps loan **Screens carefully** Default rate: **8%**		Easy to securitize Originator sells loan **Screens less** Default rate: **10–12%**

10–25% higher default rate just above the cutoff—not because borrowers are different, but because the originator's incentive to screen changes at the threshold

Better prediction would not have fixed this. The information to screen existed. The incentive to act on it did not.

What AI Changes About Moral Hazard

AI addresses moral hazard only when it changes the contract structure—not just the monitoring.

Where AI breaks through

Pricing behavior in real time

- Progressive Snapshot: telematics cut risky-driver claim costs ∼30%
- Tesla insurance prices each mile based on driving style
- The contract itself changes—premium is now a function of behavior

Where it cannot

Detection without enforcement

- Pre-2008: NLP on loan docs, covenant screening, anomaly detection
- All reduced detection cost; none prevented the crisis
- The information existed—nobody acted because incentives pointed elsewhere

Moral hazard is an incentive problem. AI helps only when it rewrites the contract, not when it improves the surveillance.

Microstructure → Risk Pricing

Kyle (1985); Glosten & Milgrom (1985); Diamond (1984)

The mechanism that discovers prices is itself expensive, noisy, and adversarial—and AI is eating this part of finance fastest.

The friction, by the numbers

- ∼80% of corporate bonds trade less than once a month; quoted prices are stale
- A $10M equity block costs 50+ basis points in market impact
- Core problem: adverse selection—market makers widen spreads when they suspect private information

Who is doing what, right now

- Citadel Securities, Virtu: ML classifies every incoming order for toxicity in microseconds
- MarketAxess Open Trading: 1,700+ counterparties, AI matching, ∼40% of IG bond volume
- Jane Street, Hudson River: deep learning models for flow prediction and inventory management
- Fragmentation persists: 16 US stock exchanges, many dark pools, the “price” depends on venue

Worked Example: Selling $10M of a Corporate Bond

Traditional: call three dealers

Dealer A: bid 98.5 · takes $3M

Dealer B: bid 98.0 · takes $5M

Dealer C: no bid today

Cost: $150–200K
$2M still unsold

AI platform: all-to-all matching

Broadcast to 1,700+ counterparties

ML matches with natural buyer (insurer)

Executes at 99.2 · full $10M

Cost: $80K
Saving: $70–120K

The equilibrium catch: as adoption grows, the pricing advantage of the platform shrinks. Spreads compress—good for investors, but the edge that made the platform valuable erodes. This is the microstructure version of the arms race.

What AI Changes About Microstructure

The strongest prediction gains in finance—and the fastest adversarial response.

Where AI breaks through

Better prediction of informed flow

- ML flags toxic orders before adverse selection hits (Citadel Securities, Virtu use this at scale)
- MarketAxess Open Trading: ML finds natural buyers for illiquid bonds that dealers won't warehouse
- Effective equity spreads fell $\sim$50% between 2000–2010 as algorithms scaled

Where it cannot

Equilibrium eats the edge

- Competitors reverse-engineer signals within months
- May 2010 flash crash: correlated algorithms withdrew liquidity simultaneously
- The faster the prediction, the faster the compression

The clearest preview of Lecture 2: gains are real, but the equilibrium response is fastest here.

What the Models Actually Do

Three live demonstrations

What the AI Tools Your Team Uses Actually Do

Mullainathan & Spiess (2017)

ML optimizes prediction, not explanation. Three live finance examples where this distinction matters.

Credit-screening vendor: "95% accurate default prediction"

- The model learns which variables correlate with default, not which *cause* it
- Happy to use ZIP code, device fingerprint, or time-of-application as proxies
- Better accuracy on paper; failure modes harder for risk and compliance to predict

Equity ML signal: factor interactions beat linear factor models

- Neural nets find momentum-by-volatility interactions linear models miss (GKX)
- The coefficients tell you nothing about why the signal works or when it stops
- Regularization makes it stable in sample but you cannot audit the mechanism

LLM summarizing an earnings call: "sentiment is positive"

- The label is correct but tells you nothing about sustainability or source of profit
- We will see this directly in a demo in a few slides

Every vendor number was optimized in-sample. The question is always: does it survive out of sample?

What LLMs Are Good At — And What They Aren't

(as of early 2026) ***LLMs are next-token predictors trained on text. A real capability, and a specific one. Finance applications succeed or fail on whether the task is in the capability.***

What they do well

- Read, summarize, extract from unstructured text
- Classify documents and tone against fixed templates
- Retrieve and recombine patterns from training data
- *Finance wins:* parse 10-Ks, tag KYC docs, summarize earnings calls, triage news, code assist
- With tools (calculator, SQL, retrieval): execute structured workflows reliably

Where they struggle

- Arithmetic and novel numerical reasoning (predict tokens, not compute)
- Long-context consistency — facts drift over 50+ pages
- Causal reasoning — they model correlation in training text
- Out-of-distribution tasks — regime shift, new jargon, rare events
- *Finance losses:* DCF from scratch, novel derivative pricing, crisis-regime forecasting

The finance question is not "can an LLM do finance?" — it's "which slice of the workflow is text interpretation (solved), numerical reasoning (solved with tools), or causal inference (still human)?"

Why Two "Equally Good" Models Disagree

Ask the same LLM the same question twice and you may get different answers. Ask two LLMs, you definitely will. Same accuracy on benchmarks. Different economic conclusions.

Same earnings call transcript. Three models. Three summaries.

Model A (GPT-4, temp 0.2)

"Strong quarter, modest guidance. Core growth durable."

$\rightarrow$ buy recommendation

Model B (Claude, temp 0.7)

"Profit beat driven by one-time tax benefit. Organic growth flat."

$\rightarrow$ hold

Model C (fine-tuned)

"Margin compression masked by cost cuts. Risk to 2026 guidance."

$\rightarrow$ reduce

All three pass a standard benchmark at $\sim$80% accuracy. All three give economically different recommendations.

Why this happens

Temperature, sampling, fine-tuning data, system prompts, context truncation — all hidden from the user.

Classical ML vs. LLMs

Two logistic regressions or two NNs can disagree too — but you can inspect coefficients, run SHAP, test feature importance. LLM disagreement is mostly a black box: you cannot decompose *why*.

The finance question

Benchmarks can't pick between models. Which model's framing ends up in the audit log?

Demo

Three live runs

The Question We Are About To Test Live

Every vendor claim reduces to one question: did the model learn the underlying concept, or the surface features of its training environment?

The experiment: fine-tune a transformer on formal financial news with sentiment labels. Evaluate on held-out news and on financial tweets covering the same firms.

The interpretation: if it learned *sentiment*, both evaluations work. If it learned *the register of formal news*, only the news does. ML prediction is always optimized on the distribution you trained on, not the underlying concept — the demo will show which one the model grabbed.

Next slide: the tool. Then we run it.

How the Model Actually Works

The tool we are testing with.

DistilBERT, 66M parameters

- Distilled version of BERT; same family as GPT/Claude, $\sim$1000$\times$ smaller
- Pretrained on Wikipedia + BookCorpus

Tokenization

- Text $\rightarrow$ $\sim$30K sub-word tokens $\rightarrow$ 768-dim vectors
- Attention layers process the sequence

What fine-tuning does

- Keep the backbone, attach a 3-class classifier head
- Gradient descent on cross-entropy loss
- 3 epochs $\times$ 2,400 articles from FinGPT sentiment dataset

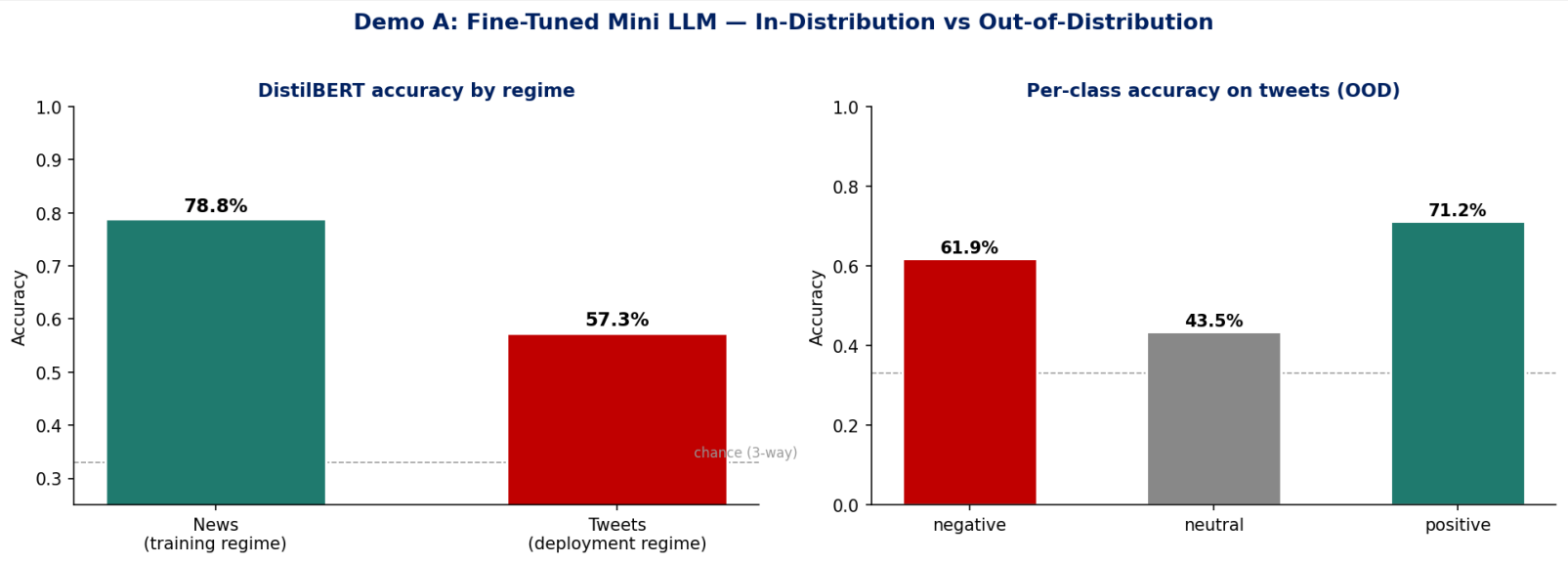
What we are *not* doing

- Not training from scratch; not using GPT-5
- No temperature—classification is deterministic argmax

HuggingFace Trainer, $\sim$3 min on a Colab T4 — the audience watches it happen live.

Evaluation: held-out news, then financial tweets.

Demo Results: Regime Shift



In-distribution: 78.8% DistilBERT fine-tuned on news. Solid.

Out-of-distribution: 57.3% Same model, tweets. 22-point collapse. And neutral sentiment almost at chance.

The architecture is not the problem—this is the same family as production LLMs. The problem is that a model trained on one distribution of financial text will fail on another.

Demo: Train Before the Crisis, Deploy Twenty Years Later

Fine-tune a DistilBERT on SEC 10-K sentences from before the GFC. Test on the post-GFC recovery, then on the modern era. Does a language-based sentiment model survive regime change?

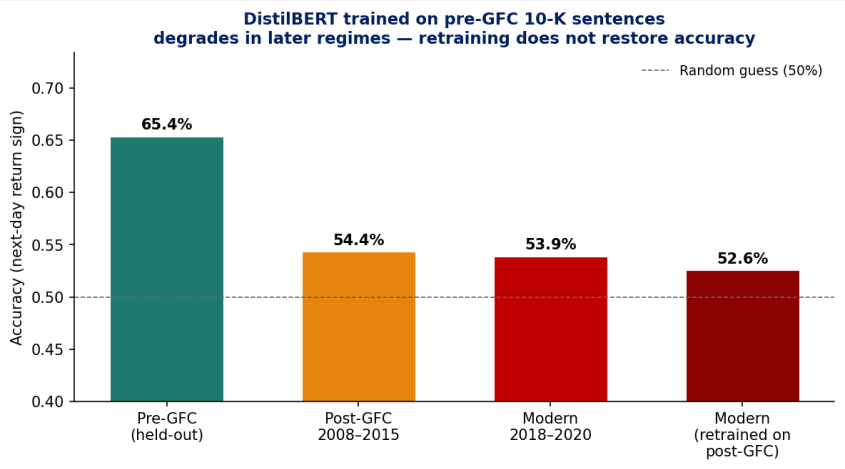
Setup

- Data: 10-K filings, *JanosAudran/financial-reports-sec* (200k sentences, 1994–2020)
- Model: DistilBERT (66M params) fine-tuned on 4,000 sentences
- Label: sign of firm's next-day return — weak supervision from market reaction
- Training: 1994–2007H1 (pre-GFC)

Three evaluation periods

- In-distribution: held-out pre-GFC sentences
- OOS post-GFC: 2008–2015 (Dodd-Frank risk disclosures, new boilerplate)
- OOS modern: 2018–2020 (cybersecurity, ESG, pandemic language)

Demo Results: The Sentiment Model Degrades Across Eras



Pre-GFC held-out: 65.4%

Real signal learned.

Post-GFC 54.4%; Modern 53.9%

Barely above random.

The language of financial disclosure changed. What moved markets in 2005 is not what moves them today.

Can Retraining Fix It?

The natural response

"Just retrain on newer data." Fine-tune a second DistilBERT on post-GFC 10-Ks (2008–2015) and test on the modern period.

What happens

- Pre-GFC model $\rightarrow$ modern: 53.9%
- Post-GFC model $\rightarrow$ modern: 52.6%
- Retraining on 2008–2015 does *not* restore modern accuracy

Why not?

The modern regime has drifted from *both* prior eras. Fine-tuning only helps on data you already have.

The harder lesson

A model retrained through a crisis learns from the crisis *and extrapolates it*. Non-stationarity is the binding constraint, not model capacity.

You can fine-tune on fresh data. You cannot fine-tune on data that does not yet exist.

What Does the Model Do With Modern Text?

We fed the pre-GFC-trained model six synthetic one-line 10-K sentences: three in pre-GFC style, three full of COVID / WFH / cybersecurity / ESG vocabulary it has never seen. What does it predict?

Pre-GFC style controls

- "principal sources of liquidity . . . cash flows" → NEG 0.67
- "capital resources adequate . . . " → NEG 0.69
- "strong demand . . . international markets" → NEG 0.68

COVID / WFH / modern vocabulary

- "COVID-19 . . . shift to remote work" → NEG 0.69
- "supply chain disruptions . . . cybersecurity" → NEG 0.69
- "ESG initiatives . . . post-pandemic" → NEG 0.68

The model does not know what it does not know. It returns the same confident NEG for pre-GFC prose and COVID-laden text alike; it hedges to the majority class and reports high confidence. There is no "I don't know" signal.

When Does This Matter? And When Does It Not?

This is not an argument against LLMs in finance. It is an argument for matching the deployment to the task's structural stability.

Low-risk (mechanics stable)

- KYC / AML document parsing
- Invoice categorization
- Extraction from stable templates (SWIFT, ISDA)
- Summarization of non-cyclical businesses

Structurally fragile

- Macro sentiment, business-cycle forecasting
- Credit risk and default across cycles
- Models reshaped by regulation (Dodd-Frank, ESG, cyber)
- Models where a crisis redefines vocabulary

Deeper point: credit risk in 2005 and 2020 is the same *economic* concept, but 10-Ks describe it with different vocabulary. A model that measures vocabulary is mis-measuring a stable concept.

Demo: Prediction vs. Understanding

Can a model that predicts sentiment distinguish between economically different sources of good news?

Two headlines, both labeled "positive"

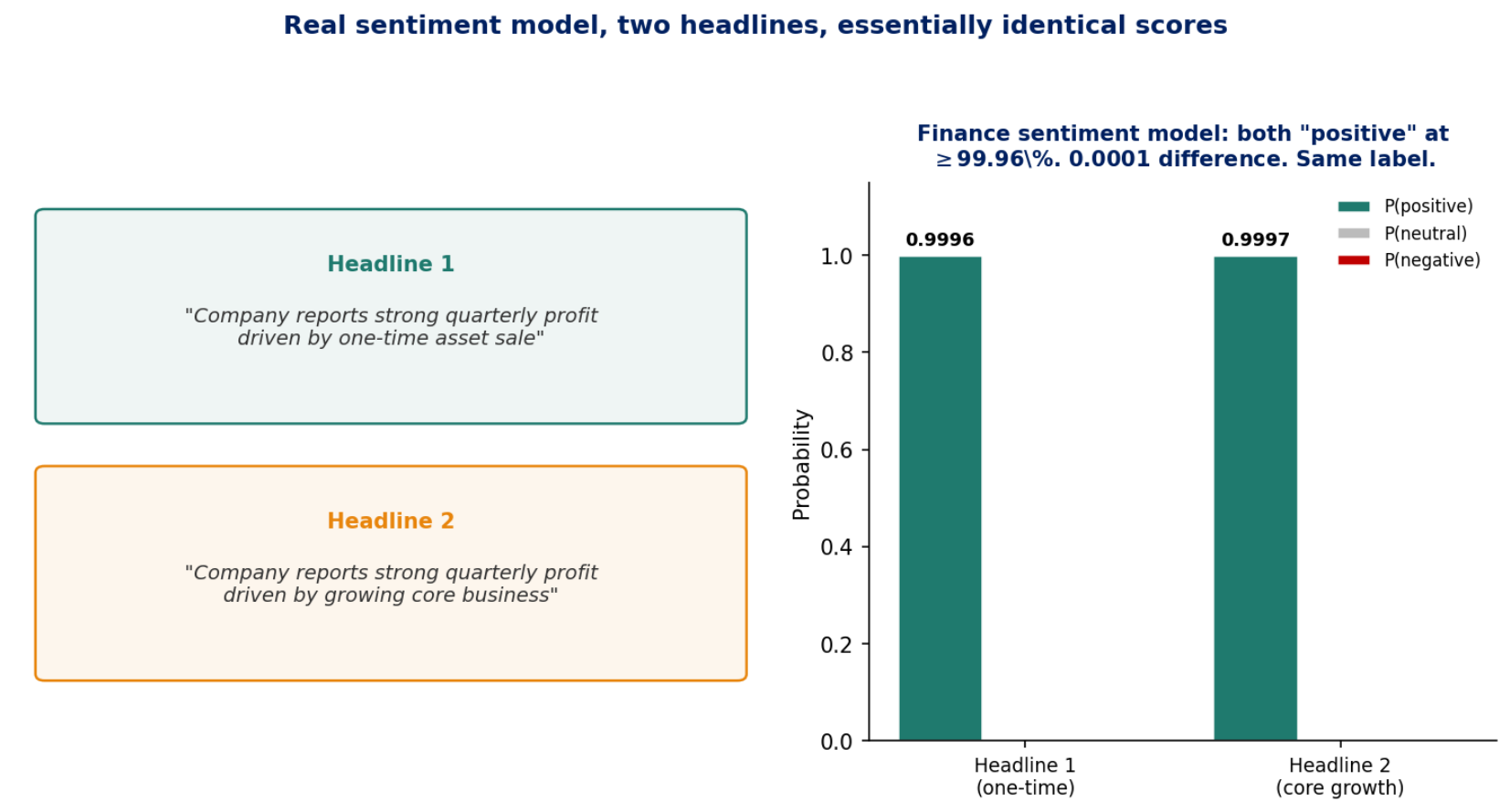
"Company reports strong quarterly profit driven by one-time asset sale"

"Company reports strong quarterly profit driven by growing core business"

Same sentiment. Different economics.

- One is sustainable; the other is non-recurring
- A credit analyst, equity investor, and regulator would treat them very differently
- Does the model?

Demo Results: Prediction $\neq$ Understanding



Identical sentiment scores for both headlines. Prediction tells you *that*; inference tells you *why*. Finance decisions require both.

Two Ways Prediction Falls Short in Finance

Across the three demos, two distinct failure modes. One about prediction quality; the other about the limits of prediction itself.

1. Prediction can be done poorly

- *Demo A* — style regime: trained on news, 79% in-distribution; 57% on tweets. Model learned the register, not the concept
- *Demo B* — temporal regime: pre-GFC model 65% $\rightarrow$ 54% modern; retraining on 2008–15 doesn't restore 2018–20
- *Root cause*: the model optimizes on the training distribution, not the underlying economic concept. Non-stationarity is the binding constraint

2. Prediction can be done well and still not be enough

- *Demo C* — prediction vs. inference: finance-tuned sentiment model gives 0.9996 vs. 0.9997 positive on "one-time asset sale" vs. "core growth"
- Accuracy is high. The economically relevant distinction is not in the label
- *Root cause*: finance decisions need the *mechanism* (why), not the outcome label (what). Prediction answers *that*; inference answers *why*

So: when is prediction alone enough? Next slide gives the framework.

When Is Prediction Enough?

Kleinberg et al. (2018)

Prediction sufficient

- Credit approval: Upstart, Zest AI
- Fraud/claim triage: Lemonade
- KYC classification: ComplyAdvantage

You do not need to know why.

Inference required

- Loan pricing under fair-lending rules
- Portfolio construction, factor timing
- Explaining a rejection to a regulator

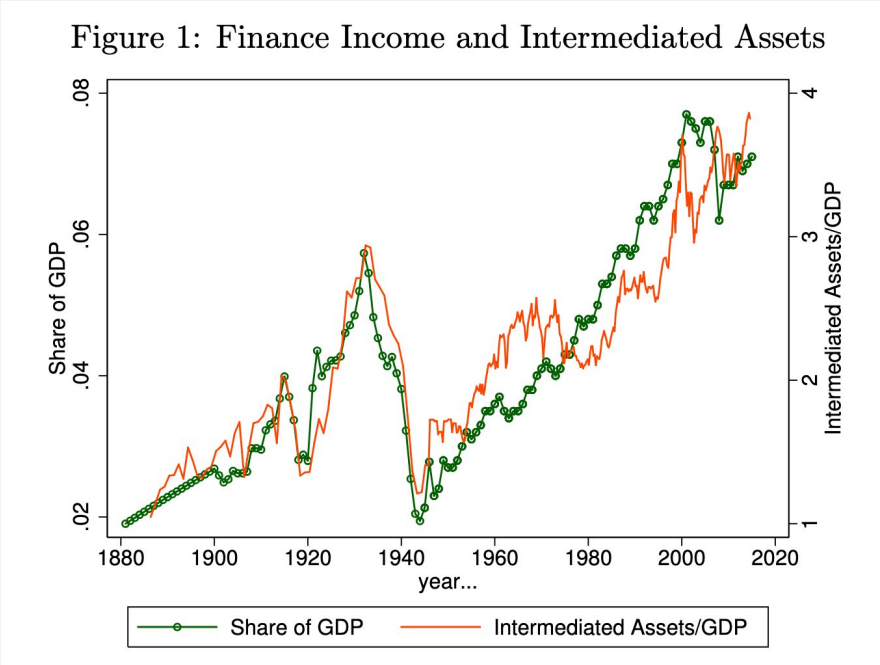
You need the mechanism, not just the pattern.

Kleinberg conditions for welfare gains: (1) decision depends on a forecast; (2) forecast is improvable; (3) improvement survives real-world constraints.

The Track Record

What Does Intermediation Actually Cost?

Philippon (2015, 2019)

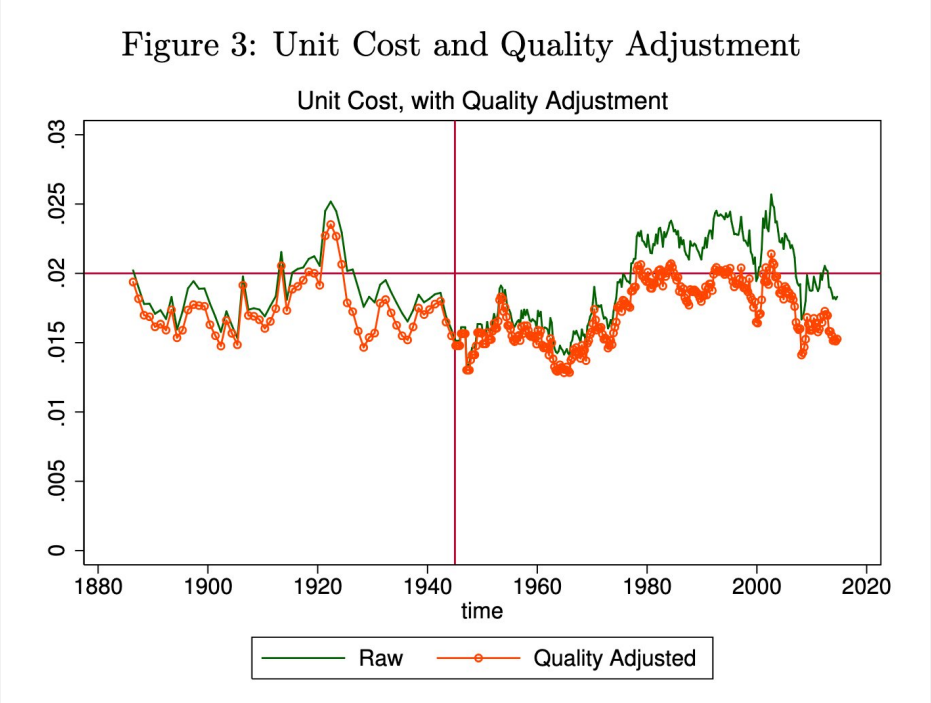


Reading this figure

- **Green:** what society pays the finance industry (share of GDP)
- **Orange:** volume of assets being intermediated
- Both have grown enormously over the past century
- **But their ratio—the unit cost of financial intermediation—has not changed**

130 Years of Financial Technology

Philippon, Figure 3



Quality-adjusted unit cost: 1.5–2% for 130 years.

- ATMs did not bend this curve
- Electronic trading did not
- Internet banking did not
- Zero-commission brokerage did not

This is not anti-technology. It is a discipline for evaluating claims about the current wave.

Where Did the Real Gains Go?

A flat unit-cost line does not mean no welfare gain. Technology does improve financial services — but the gains flow through three channels, and only one of them shows up in Philippon's measure.

1. Intermediaries capture the gain *Philippon (2015) rent-extraction view*

Non-competitive segments — prime brokerage, OTC derivatives, structured products, high-end advisory — retain the surplus. Cost to produce the service falls; price to the client does not. Unit cost visible in Philippon's data: unchanged.

2. Shared with borrowers in competitive segments *Greenwood & Scharfstein (2013)*

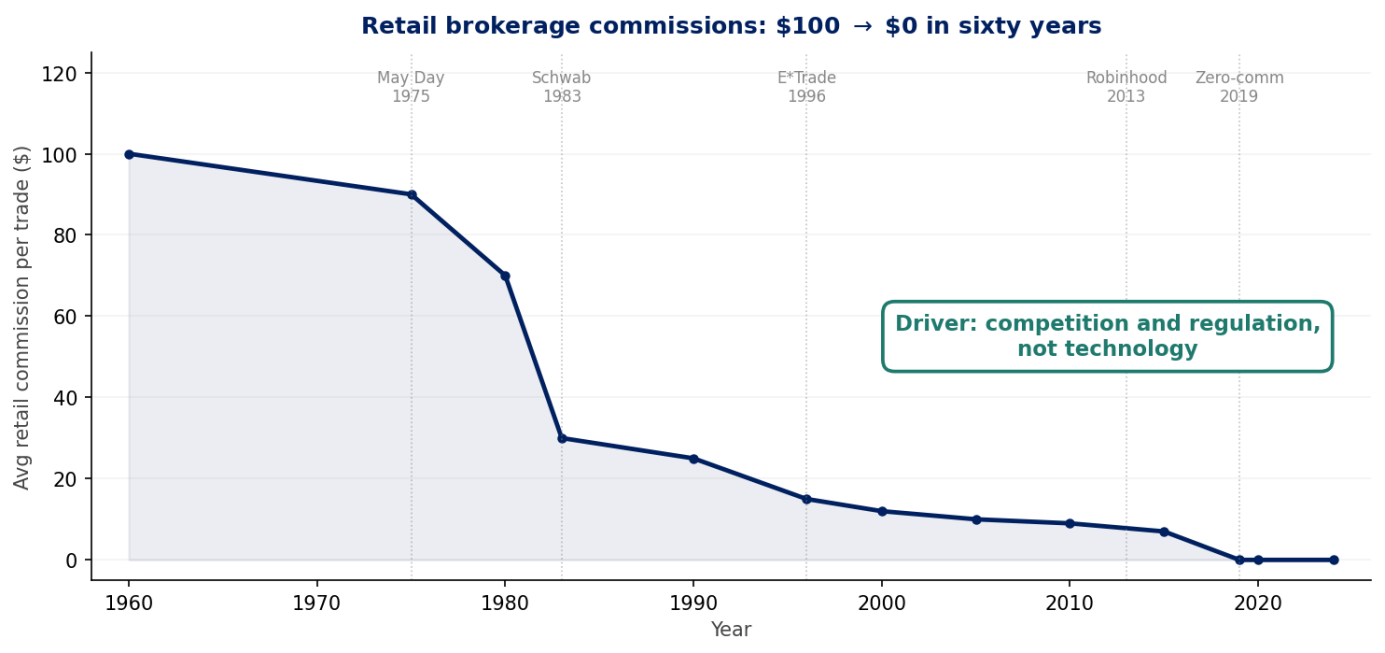
Retail brokerage commissions, mortgage refi, deposit rates, index-fund fees: competition forces partial pass-through. Unit cost in that vertical does fall — and Philippon sees it, partly offsetting gains in other segments.

3. Pass-through via access *Philippon (2016) "FinTech Opportunity"*

Fintech underwriting, unbanked enrollment, robo-advisors for households that could not afford an RIA. Gains go into *serving new customers* at roughly the old unit cost. Philippon's average does not move. Welfare rises anyway.

The test for AI is not "did unit cost fall?" but "where is the gain flowing — and are we measuring it?"

Where Cost Declines Actually Came From



Asset management tells the same story

- Active funds: 1–2% annual fees with persistent underperformance
- Index fund share grew from ∼10% to over 50%
- Drivers: Vanguard's scale economics, accumulating academic evidence, fee transparency

Overfitting: Every Backtest Is Optimistic

Hastie, Tibshirani & Friedman (2009); Harvey, Liu & Zhu (2016)

Overfitting

Vendor: "95% accuracy"

Out of sample: $\sim$52%

- Trained and tested on the same data
- Memorized noise as if it were signal

Data snooping

1,000 random strategies

- $\sim$50 "pass" significance at $p = 0.05$
- 15 show Sharpe > 1.0
- **All of them are pure noise**

Survivorship bias

Failed firms disappear from the dataset

Look-ahead bias

Using information not available at decision time

Non-stationarity

Training regime is not the production regime

This is the universal pattern: every strategy looks better in the backtest than in production.

The Arms Race: A History

Statistical arbitrage (1990s–2000s)

- D.E. Shaw, Renaissance: pairs trading, factor long-short. Early Sharpe ratios >2
- AUM in quant equity grew from $\sim$\$100B (2000) to $\sim$\$500B (2007)
- Returns declined as capital poured in and strategies converged
- August 2007: quant crisis—\$1.4T in correlated positions unwound in days

The lesson: the edge was real, but temporary. Adoption was the mechanism that destroyed it.

High-frequency trading followed the same arc

- Reduced spreads dramatically in the 2000s—genuine value for investors
- HFT firms' own profits compressed as competition intensified and costs rose

The Current Wave: LLM Sentiment

Lopez-Lira & Tang (2023)

What Lopez-Lira & Tang actually show

- GPT-3.5 asked to label news headlines as positive, negative, or neutral, Oct 2021–Dec 2022
- A long-positive / short-negative strategy earns large out-of-sample returns
- Effect is stronger on smaller firms and on simpler news items
- Paper is a *demonstration of predictive power*, not a test of long-run persistence

Discovery → **Adoption** → **Crowding** → **Compression**

We're at "Discovery." The historical record (stat arb, HFT) suggests Adoption and Crowding follow, and Compression is what happens as everyone uses the same signal. Lopez-Lira & Tang document the first step; the remaining three are not yet tested empirically for LLM sentiment.

The Zillow iBuyer Case: Market-Making in a Bad Asset

Buchak, Matvos, Piskorski & Seru (JPE, forthcoming) ***Zillow Offers was an attempt to dealer in houses — the structural economics (Kyle / Glosten-Milgrom) predict the failure; Zillow's execution mistakes made it worse.***

Zillow's own explanation (Nov 2021)

- "Home-price forecasting had become too unpredictable"
- Scaling would create too much balance-sheet volatility
- Renovation and resale capacity constraints
- Q3: $304M writedown; total ∼$881M; 25% staff cut

Zillow-specific execution

- Scaled aggressively into the 2021 housing peak
- ∼9,700 homes on balance sheet Q3 2021; couldn't clear
- Operational bottleneck in renovation and resale

The structural problem (Buchak et al.)

- Value prop is *speed* — instant offer, no inspection
- Speed requires automated valuation (less property-specific info)
- Seller knows more than fast algorithmic buyer: adverse selection
- iBuyers do best on homes already easy to price — where extra liquidity is least needed

What survives, and why

- Opendoor, Offerpad operating (2026) — smaller, asset-light
- Zillow exited 2021; Redfin exited 2022
- Survivors shifted to tighter turns, lower capital intensity

Houses are market-maker-hostile: heterogeneous, illiquid, expensive to inspect, asymmetric information between seller and algorithmic buyer. ML valuation doesn't fix that.

What We Have Established

Risk is the central problem

- Markets measure, price, transfer risk
- AI changes the tools, not the frictions

The historical record

- 130 years of flat intermediation cost
- Every profitable strategy was competed away

Prediction is necessary but not sufficient

- ML improves prediction in many settings
- Value depends on decision, constraints, who captures surplus

The right question

- Which friction, which mechanism?
- Out of sample? At scale?

What Lecture 1 Does Not Answer

What happens at scale?

- When every firm adopts the same ML models, system-level effects emerge
- Crowding, alpha decay, endogenous changes to the data-generating process

Can we trust AI as an economic agent?

- LLMs do not just provide inputs—they frame problems, make recommendations, take actions
- Their systematic biases become institutional biases

Does AI create entirely new markets?

- This lecture asked whether AI improves existing financial activities
- When AI makes the previously unobservable observable, new contracts and securities become feasible

These are the questions for Lecture 2.

Better prediction is necessary

but not sufficient. *The economic value depends on the*

decision,
the constraints, and the equilibrium response.